

Study on the next data frontier: generative AI, regulatory compliance and international dimensions

Executive summary

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October 2025

EUROPEAN COMMISSION

Directorate-General for Communications Networks, Content and Technology

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PDF ISBN 978-92-68-31932-1 doi:10.2759/3909870 KK-01-25-135-EN-N

Luxembourg: Publications Office of the European Union, 2025

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Executive Summary

This study, commissioned by the Directorate-General for Communications Networks, Content and Technology (DG CNECT) of the European Commission, explores three critical dimensions of the evolving European data economy. Task 1 examines data needs for innovation and the role of synthetic data; Task 2 analyses the use of automated data reporting mechanisms for regulatory compliance across key sectors; and Task 3 addresses international aspects of the data economy. The methodology combines desk research, targeted surveys, stakeholder interviews, and comparative analysis, drawing on academic and policy literature, market studies, and direct consultation with industry and sectoral experts to ensure robust, evidence-based findings.

Task 1: Data needs for innovation and the role of synthetic data

Europe's Generative AI (GenAI) ecosystem is growing but remains behind the United States and China in foundational model development, investment, and market share. The study identifies Europe's strengths in research output and innovation, supported by a vibrant academic and start-up environment. However, most European GenAI companies focus on application layers—such as text, image, and speech tools—rather than building foundational models, which require significant compute resources and financial investment.

Access to high-quality, multilingual, and domain-specific datasets emerges as a critical bottleneck for innovation. Scientific literature and stakeholder feedback highlight that foundational models require vast amounts of unstructured data; while fine-tuning and specialised applications depend on smaller, highly curated datasets. Linguistic and cultural diversity is essential, as the dominance of English in training data risks underperformance in other languages and cultural underrepresentation.

Legal, technical, and practical barriers persist, particularly for SMEs. GDPR compliance, copyright restrictions, and fragmented enforcement across Member States increase complexity and costs. Technical challenges include the scarcity of annotated, multilingual datasets and risks of bias in training data. Stakeholders report difficulties in accessing domain-specific and multimodal datasets, compounded by high licensing fees and reluctance among data holders to share assets due to trust and competitive concerns.

The cost of acquiring, preparing, and managing data is a major operational burden. Licensing fees, annotation services, and compliance costs dominate expenditure for European GenAI firms. Public initiatives such as Common European Data Spaces and GAIA-X are widely anticipated to improve data accessibility, but awareness and participation remain limited, especially among SMEs.

Task 2: Automated data reporting for regulatory compliance

Automated data reporting tools are increasingly relevant for demonstrating regulatory compliance in agriculture, automotive, healthcare, chemicals, and energy. These sectors face complex and evolving EU regulations, including CAP Strategic Plans (agriculture), MDR/IVDR (healthcare), REACH and CLP (chemicals), REMIT and ETS (energy), and sustainability directives (automotive). Despite the growing availability of digital tools, adoption remains uneven. Standard methods—such as spreadsheets and manual data entry—continue to dominate, particularly among SMEs, due to their low cost and flexibility, even though they are labour-intensive and error-prone.

Advanced tools, including digital logbooks, Farm Management Information Systems (FMIS), ERP systems, and specialised compliance software, are gaining traction among larger organisations. These solutions enable structured data capture, integration with geospatial and sensor data, and improved traceability. However, uptake is constrained by high licensing costs, technical complexity, and interoperability issues with national portals and

legacy IT systems. Fully automated solutions, such as blockchain-based traceability platforms, remain experimental and limited to pilot projects.

Barriers to adoption include financial costs, digital skill gaps, and fragmented regulatory frameworks. Smaller operators face disproportionate challenges, risking exclusion from increasingly digitalised compliance regimes. Cost-benefit assessments indicate that advanced digital logbooks offer the best balance of efficiency and affordability, while FMIS and fully automated tools promise transformative benefits but remain viable primarily for large enterprises.

Investment in interoperable infrastructure, harmonised standards, and digital skills is essential. Partnerships and open data ecosystems should enable access to quality datasets, while flexible regulation and SME support ensure inclusive, scalable digital transformation.

Task 3: International aspects on the data economy

The analysis of international approaches to data governance across eight key economies identifies pathways for the EU to shape a more open, secure and rights-respecting global data ecosystem. Convergence with EU regulation varies markedly across the targeted countries. Japan's 2021 APPI and Data Free Flow with Trust concept achieve full alignment with EU standards, while Canada and South Korea hold adequacy decisions under GDPR but still face sectoral fragmentation. Australia and Brazil exhibit partial convergence through pragmatic, business-friendly models. The UK retains EU adequacy post-Brexit yet risks divergence under its Data (Use and Access) Act. The United States follows a market-driven, sector-specific patchwork, and China enforces a state-driven model emphasising localisation and security controls.

Four governance paradigms coexist: rights-driven (EU GDPR), market-driven (US innovation), state-driven (Chinese sovereignty) and consensus-driven (Japan's hybrid model). The EU's "third way" seeks to combine robust individual protections with innovation-friendly, federated data architectures.

Six illustrative use cases reveal practical opportunities and common barriers: 1) Digital Battery Passports link EU, North American and Asian manufacturers under the EU Battery Regulation; 2) Federated Learning for Rare Disease Genomics shows how collaborative AI can advance health research without sharing patient data; 3) Interoperable Safety Data Exchange for Autonomous Vehicles demonstrates the need for neutral "data trustees" to aggregate incident data; 4) Deforestation-Free Supply Chain Traceability under the EU Deforestation Regulation calls for inclusive, low-cost traceability platforms for smallholders; 5) Privacy-Enhancing Passenger Name Record (PNR) exchange offers a rights-respecting security model; and 6) Federated Digital Twins for Aerospace highlights both industrial promise and IP-protection challenges.

To capitalise on these findings, the EU should leverage its regulatory gravity – for example through sectoral adequacy agreements and Gaia-X/IDSA-compliant pilot projects – to export interoperable data spaces; pioneer novel governance models, such as data trustees and privacy-enhancing technologies, to address persistent trust deficits; forge practical partnerships (for instance a "Traceability for Sustainable Trade" initiative with Brazil) to turn regulatory friction into cooperation; prioritise PETs in security cooperation and establish clear definitions for sensitive and highly sensitive non-personal data; and scale up R&D funding in generative AI, federated learning and cross-border data infrastructure. By aligning robust safeguards with technical and governance frameworks, the EU can solidify its role as the global standard-setter for a human-centric, innovation-driven data economy.



Publications Office
of the European Union

doi:10.2759/3909870

ISBN 978-92-68-31932-1